Extra Practice Multiplying and Dividing Rational Expressions

MCR3U

Jensen

1) Multiply the following rational expressions. Simplify and state restrictions.

a) $\frac{x+4}{x} \cdot \frac{x^2}{x^2+5x+4}$

b) $\frac{x^2+10x+16}{5x-10} \cdot \frac{x-2}{x^2+9x+8}$

c) $\frac{2x^2-10x}{x^2-9x+20} \cdot \frac{x^2-8x+16}{4x^2}$

d) $\frac{2x+4}{x+4} \cdot \frac{5x^2+21x+4}{10x+2}$

2) Divide the following rational expressions. Simplify and state restrictions.

a) $\frac{x^2-5x+6}{5} \div \frac{x-3}{15}$

b) $\frac{x^2-5x+6}{8x^2+24x} \div \frac{x-2}{4x+12}$

c) $\frac{x^2-2x+1}{x+1} \div \frac{x^2-1}{x+1}$

d) $\frac{x^2+7x+12}{x^2+3x-10} \div \frac{x^2-x-20}{x^2-25}$

Answers

1) a) $\frac{x}{x+1}$; $x \neq -4, -1, 0$ **b)** $\frac{x+2}{5(x+1)}$; $x \neq -8, -1, 2$ **c)** $\frac{x-4}{2x}$; $x \neq 0, 4, 5$ **d)** $x + 2$; $x \neq -4, -\frac{1}{5}$

2) a) $3x - 6$; $x \neq 3$ **b)** $\frac{x-3}{2x}$; $x \neq -3, 0, 2$ **c)** $\frac{x-1}{x+1}$; $x \neq -1, 1$ **d)** $\frac{x+3}{x-2}$; $x \neq -5, -4, 2, 5$